

Article

Oil Palm Fruit Maturity Classification Algorithm Based on LAB Dual-Branch Feature Fusion

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Featured Application

The proposed method can be applied to non-destructive oil palm fruit maturity assessment, automated grading, and intelligent harvesting decision support.

Abstract

Accurate classification of oil palm fruit maturity is critical for optimizing harvest schedules and maximizing yield. To address the limited discrimination accuracy caused by subtle color variations, this paper proposes an LAB dual-branch feature fusion method for five-class oil palm fruit maturity classification. A classification model named CMF-Net is constructed, in which the luminance (L) and chrominance (ab) components are decoupled into two parallel branches to separately extract structural and color information. An input adaptation module is introduced to perform channel mapping and feature enhancement. During feature fusion, a gated fusion mechanism is employed for adaptive integration of the dual-branch features, followed by a BiFPN module for multi-scale feature interaction. A multi-level attention enhancement module composed of coordinate attention and self-attention is designed to strengthen key-region perception and global contextual modeling. An auxiliary color regression constraint and ordinal supervision are incorporated to improve the model's sensitivity to subtle chromatic changes and the progressive relationship among maturity stages. The proposed algorithm provides a color-decoupled and order-aware representation framework for gradual fruit maturity classification, offering a useful reference for fine-grained agricultural visual classification tasks involving continuous color transitions. Experimental results on a public dataset show that CMF-Net achieves a validation accuracy of 94.66%, outperforming mainstream models by 1.3–6.1 percentage points, while also yielding superior recall and F1-score.



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Keywords: oil palm fruit; maturity classification; LAB color space; dual-branch network; feature fusion; attention mechanism; ordinal learning

1. Introduction

As one of the most productive oil-bearing crops globally, the oil palm fruit features an oil content that correlates strongly with its ripeness. Across Southeast Asia, Africa, and Latin America, it stands as a vital economic pillar, supporting local agricultural livelihoods while ensuring the stability of global edible oil supplies. In palm oil processing, maturity control mainly takes place during the raw material handling stage before sterilization.

Mixing fruit bunches with different maturity levels can affect oil yield and product quality. Therefore, it is important to assess fruit maturity before sterilization. By equipping automated production lines with intelligent vision systems, real-time maturity recognition and quality monitoring of fruit bunches can be achieved, providing support for process control. Without changing the existing workflow, this approach can improve oil yield and product consistency, and support the intelligent development of the palm oil industry [1].

In practical palm oil production, oil palm fruits are commonly categorized into different maturity stages according to external visual characteristics, especially exocarp color and the degree of fruit detachment [2]. Accurate identification of these maturity stages is important because adjacent stages often exhibit gradual color transitions and similar surface appearances. Early research efforts largely relied on handcrafted features; for instance, some studies utilized color features combined with Canny edge detection to achieve recognition [3], while others employed Principal Component Analysis (PCA) or Linear Discriminant Analysis (LDA) for feature dimensionality reduction, subsequently training Support Vector Machine (SVM) classifiers using Red, Green, and Blue (RGB) or grayscale images [4]. More recently, image-based maturity assessment has been extended from single-source visual features to multimodal sensing. Lin et al. [5] integrated RGB and multispectral images for tomato maturity classification and fruit counting, demonstrating the potential of complementary visual and spectral information for automated maturity monitoring in greenhouse environments.

With the rapid development of deep learning, its application in maturity classification tasks has become increasingly widespread and has gradually emerged as the dominant approach in this field. Thus, the analysis and design of maturity classification can be mainly organized into three branches. The first branch relies on CNN approaches, such as AlexNet [6,7], RetinaNet [8], DenseNet [9], MobileNetV2 [10], MnasNet [11], EfficientNet [12], and CNN models combined with transfer learning and data augmentation strategies [13]. These methods mainly focus on extracting discriminative visual features for image-level maturity classification. Owing to real-time performance and high accuracy of YOLO, the second branch relies on YOLO methods, such as YOLO-ELS [14], MTD-YOLO [15], and DCS-YOLO [16], which are commonly used when fruit localization and maturity recognition need to be performed simultaneously. The third category is based on vision transformers (ViTs), which represent an important shift in visual recognition by enabling long-range dependency modeling and improved scalability for complex visual tasks [17]. However, training Transformer-based models with strong performance and good generalization ability usually requires higher computational resources than conventional CNNs. Therefore, in practical agricultural vision applications, Transformer-related models can be broadly divided into pure Transformer-based models and CNN–Transformer hybrid models. For example, Rai et al. [18] proposed FruitProm, a Transformer-based maturity estimation and detection framework for fruits and vegetables by introducing a probabilistic maturity prediction head into RT-DETRv2. Amad et al. [19] presented a two-step semi-supervised approach that leverages knowledge distillation and transfer learning for citrus maturity detection in uncrewed aerial vehicle images. Khan et al. [20] proposed a convolutional Transformer framework for tomato maturity recognition, combining convolutional feature extraction with Transformer-based contextual modeling. In addition, Yu et al. [21] proposed a quality-driven Transformer–CNN hybrid model for the fusion of external and internal fingerprint features, further demonstrating the effectiveness of hybrid Transformer–CNN feature fusion in complex visual recognition tasks. These studies indicate that Transformer-based architectures have shown promising potential in fruit maturity and quality assessment tasks, but their computational cost and data requirements should also be considered in practical agricultural deployment.

More recently, several novel approaches have been proposed for maturity classification, such as LAB dual-branch network. Dual-branch neural networks employ decoupled modeling of different features, achieving promising results in remote sensing image analysis [22], medical image segmentation [23], and industrial defect detection [24]. These networks can effectively extract key features in complex backgrounds and improve the model's discriminative capability. In maturity recognition tasks, color information serves as an important basis for distinguishing different growth stages. Relevant studies have demonstrated the critical role of color features in maturity determination [25]. For example, Zhao et al. [26] focused on different wheat varieties, converted RGB images to the LAB color space, and extracted the a-component to analyze color variation characteristics across different maturity stages. The results demonstrated that LAB color components can effectively reflect crop maturity variation patterns. Sabri et al. [27] addressed the maturity classification of fresh oil palm fruit bunches, comparing multiple color models including RGB, HSV, and LAB, and combined color moment features with support vector machines for classification. The results indicated that the LAB model achieved better classification performance than the RGB model, demonstrating its superior discriminative capability in maturity representation. Compared with the RGB space, LAB can separate luminance and chrominance information and exhibits stronger robustness to illumination variations. The research by Saifullah et al. [28] further validated its superiority in accuracy and stability over traditional RGB methods.

On another research frontier, attention mechanisms have been widely employed to enhance the model's capability in modeling fine-grained information. These mechanisms apply weighted attention to key regions or channel features, improving the model's response to important features. For example, Herman et al. [29] introduced a residual attention mechanism in oil palm fruit maturity classification to enhance the capability of identifying subtle differences. Zhao et al. [30] employed a self-attention mechanism based on a lightweight Transformer to achieve global feature modeling, maintaining high recognition performance while reducing computational complexity. Additionally, Ji et al. [31] proposed the FMA-YOLO model, combining CBAM, coordinate attention, and EMA mechanisms to improve detection performance in complex environments.

The aforementioned methods have achieved great progress in maturity classification and detection tasks; however they still face challenges in complex scenarios. For example, fruit maturity shows continuous color changes, and adjacent maturity levels only differ slightly in color ratio, spot distribution, and texture roughness, which can cause confusion between nearby classes, and in the RGB color space, brightness and color information are mixed together, making it hard to highlight subtle color differences related to maturity changes. Furthermore, existing methods typically focus on extracting overall features but ignore the collaborative modeling of local color changes and brightness information. Effectively separating brightness and color information while enhancing collaborative modeling of local key regions remains a significant challenge, which is a key motivation for our research.

Therefore, how to effectively separate brightness and color information, and improve the ability to model key local regions together, remains a key problem in maturity classification. To address this issue, this paper proposes an oil palm fruit maturity classification model based on LAB dual-branch feature fusion with multi-level attention enhancement, namely the Color-Matrix Fusion Network (CMF-Net). The model converts input images to the LAB color space and employs an input adapter to perform channel mapping for the L luminance branch and ab chrominance branch, adapting them to the input structure of the pre-trained network. A dual-branch feature extraction network is utilized to learn luminance texture information and chrominance discriminative information separately, and

a gated fusion mechanism adaptively adjusts the contribution weights of the two feature types, achieving complementary luminance and color feature representation. The model further introduces BiFPN for multi-scale feature interaction, combining coordinate attention and self-attention mechanisms to strengthen key region localization and global context modeling capabilities, thereby improving the discriminative ability for fine-grained maturity differences. Additionally, an ordinal supervision strategy is designed to incorporate the ordinal relationships between maturity levels into the training process, enabling the model to better learn the continuous variation patterns of adjacent maturity stages. Experimental results on the main Oil palm fruit ripeness dataset and supplementary external datasets demonstrate that the proposed CMF-Net achieves promising performance across multiple evaluation metrics and shows good generalization potential.

2. Materials and Methods

2.1. Dataset Description

The main Oil Palm Fruit Ripeness Dataset used in this study consists of oil palm fruit images collected under a specific acquisition scenario. The images are categorized into five maturity levels, Immature, Partially ripe, Fully ripe, Overripe, and Decayed, as shown in Figure 1. These five categories represent a fine-grained maturity grading task, where adjacent maturity stages often exhibit subtle differences in color, texture, and surface appearance. In particular, the transition from partially ripe to fully ripe and from overripe to decayed can be difficult to distinguish because of gradual changes in fruit color and surface degradation.

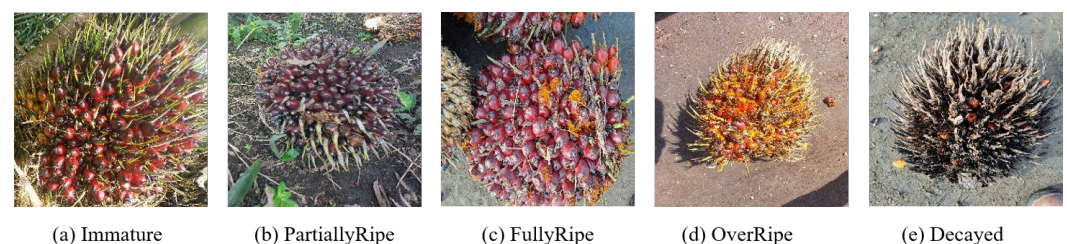


Figure 1. Five maturity levels of oil palm fruits.

To further evaluate the generalization ability of the proposed CMF-Net, three additional public datasets were introduced, and their sources, fruit types, class mappings, and sample distributions are summarized in Table 1.

Table 1. Summary of the datasets used in this study.

Dataset	Source	Fruit Type	Classes and Samples
Oil Palm Fruit Ripeness Dataset	Mendeley Data	Oil palm fruit	Immature (759), Unripe (1420), Under-ripe (1496), Ripe (667), Over-ripe (413)
FFB dataset	Kaggle	Oil palm FFB	Immature (460), FullyRipe (460), OverRipe (460)
Palm ripeness dataset	Mendeley Data	Oil palm fruit	Immature (91), PartiallyRipe (72), FullyRipe (91)
Date Fruit Dataset	IEEE DataPort	Date fruit	Immature (2323), PartiallyRipe (1727), FullyRipe (1208), OverRipe (1174)

2.2. Data Preprocessing and Augmentation

For all datasets, images were first organized according to their mapped maturity labels, and the same train–validation split was used for CMF-Net and all baseline models on each dataset to ensure a fair comparison. All images were resized to 224×224 pixels before being fed into the models. For CMF-Net, each RGB image was converted into the

CIELAB color space, where the luminance channel L was used as one input branch and the chromatic channels a and b were concatenated as the second input branch. The L , a , and b channels were normalized to the range $[0, 1]$, allowing the model to learn luminance-related texture information and chromatic maturity cues separately. This separation is particularly important for adjacent maturity stages, where the visual differences are often reflected in gradual chromatic shifts rather than obvious shape changes. For the baseline models, including ResNet152V2, MobileViT-CBAM, Swin-T, ConvNeXt-Tiny, and EfficientNetV2-S, RGB images were resized to 224×224 pixels and normalized according to the preprocessing strategy used by each model implementation. For models initialized with ImageNet pretrained weights, ImageNet-style normalization or the corresponding framework-specific preprocessing was adopted. The data preprocessing pipeline is shown in Figure 2.

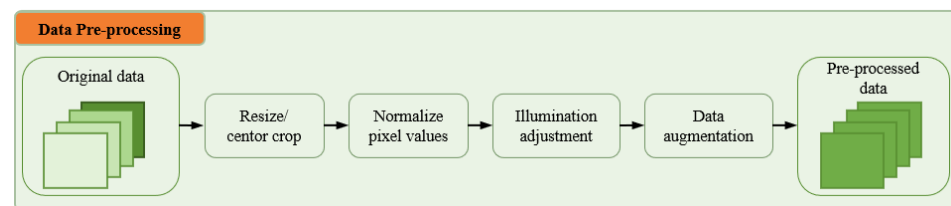


Figure 2. Data preprocessing pipeline for CMF-Net.

Data augmentation was applied only to the training set. For CMF-Net, augmentation was performed in the LAB space, including random horizontal flipping, brightness perturbation on the L channel, small-angle rotation, and Gaussian noise injection. The brightness factor was randomly sampled from $[0.5, 1.5]$, the brightness bias from $[-0.1, 0.1]$, the rotation angle from $[-15^\circ, 15^\circ]$, and Gaussian noise with a standard deviation of 0.03 was added to both the luminance and chromatic channels. Label smoothing with a factor of 0.02 was applied to the classification labels during training. For the RGB-based baseline models, random horizontal flipping, random rotation, random resized cropping, and color jittering were used during training. No data augmentation was applied to the validation set, ensuring that all models were evaluated on unmodified validation images while improving robustness during training. The augmentation procedures are shown in Figure 3.

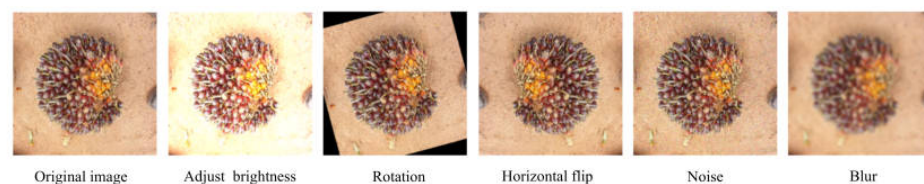


Figure 3. Data augmentation strategies for training samples.

2.3. Overall System Architecture

In this section, we present the proposed model of CMF-Net in detail. As shown in Figure 4, the model CMF-Net consists of six components: Data Pre-processing and LAB Decomposition, Dual-Branch Feature Extraction, Gated Feature Fusion, BiFPN-based Multi-scale Feature Interaction, Attention Enhancement, and Ordinal Classification. The Data Pre-processing and LAB Decomposition part transforms the original RGB images into luminance and chrominance maps. The Dual-Branch Feature Extraction part is shown in Figure 4a,b, which extracts luminance-structural features and chrominance-discriminative features through two parallel branches. The Gated Feature Fusion part is shown in Figure 4c, which adaptively integrates the features from the two branches. Then, the fused features are processed by BiFPN, coordinate attention, and self-attention modules. Finally, the Ordinal Classification part obtains the final ripeness category through sigmoid prediction and threshold decoding.

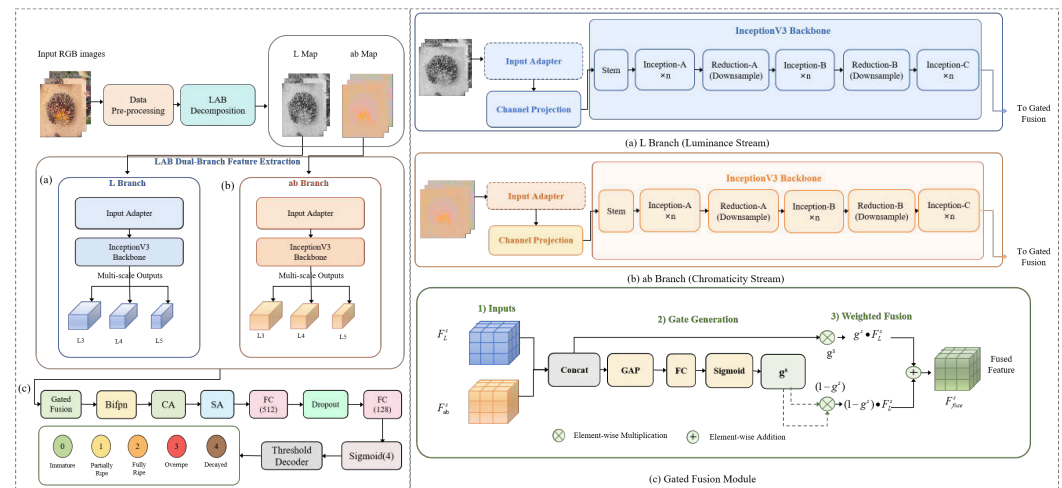


Figure 4. Structural Diagram of CMF-Net for Oil Palm Fruit maturity Classification.

In the Data Pre-processing and LAB Decomposition part, the input RGB images are first resized and normalized. Then, the pre-processed RGB images are converted into the LAB color space. Based on the LAB representation, the image is decomposed into the L map and the ab map. The L map contains one luminance channel, which mainly describes brightness-related structure and texture information. The ab map contains two chrominance channels, which mainly represent color distribution information related to oil palm fruit ripeness.

The Dual-Branch Feature Extraction part includes the L branch and the ab branch. As shown in Figure 4a, in the L branch, the one-channel L input is first projected into a three-channel representation by an input adapter. As shown in Figure 4b, in the ab branch, the two-channel ab input is also projected into a three-channel representation. The input adapter is composed of a 1×1 convolution, batch normalization, and ReLU activation. After channel projection, the two inputs are fed into two parallel InceptionV3 backbone networks. The backbone extracts multi-scale feature maps from different stages, denoted as L_3 , L_4 , and L_5 , corresponding to feature resolutions of $1/8$, $1/16$, and $1/32$, respectively.

In the Gated Feature Fusion part, the features from the L branch and the ab branch are fused adaptively. As shown in Figure 4c, the gated fusion module first concatenates the two branch features along the channel dimension. Then, global average pooling, a fully connected layer, and a sigmoid function are used to generate the gate weight. By using this learned weight, the model can adjust the contributions of luminance features and chrominance features, obtaining fused features for the following modules.

In the BiFPN-based Multi-scale Feature Interaction part, the fused features are further processed by the BiFPN module. The BiFPN module uses bidirectional feature paths to exchange information among multi-scale feature maps. In this way, low-level texture details and high-level semantic features can be combined more effectively, improving the feature representation for adjacent ripeness stages.

In the Attention Enhancement part, the fused multi-scale features are enhanced by coordinate attention and self-attention. The coordinate attention module is used to strengthen the spatial location information of key ripeness-related regions. The self-attention module is then used to model global relationships among different feature regions, obtaining enhanced feature maps with stronger discriminative ability.

In the Ordinal Classification part, the enhanced feature maps are first transformed into feature vectors by global average pooling. Then, the feature vectors are processed by fully connected layers and dropout. Considering the ordinal relationship among oil palm fruit ripeness levels, an ordinal prediction head is used to output $K - 1$ ordered

probabilities. Finally, the ripeness category is obtained through threshold-based decoding. For the five-level classification task, the model outputs immature, partially ripe, fully ripe, overripe, and decayed categories.

Remark 1. Figure 4 describes the novel CMF-Net framework proposed in this paper, including data preprocessing and LAB decomposition, LAB dual-branch feature extraction, gated fusion mechanism, BiFPN-based multi-scale feature interaction, multi-level attention enhancement, and ordinal classification. The merit of this design is that CMF-Net can more accurately classify oil palm fruit ripeness when adjacent maturity levels exhibit subtle color transitions and when luminance and chrominance information are coupled in the RGB space. By introducing the LAB color space to separate luminance and chrominance information into two branches, and by integrating the gated fusion mechanism with multi-level attention enhancement, the proposed model strengthens the synergistic representation of luminance-structural information and chromaticity-discriminative information.

2.4. Dual-Branch Feature Fusion and Enhancement Mechanism

The dual-branch InceptionV3 architecture effectively extracts image luminance structural information and chromaticity distribution features. Building upon this foundation, this paper introduces an input adaptation module to achieve a unified representation of features from different modalities, and constructs a gated fusion mechanism following the backbone network to perform adaptive weighted integration of luminance and chromaticity features. Subsequently, the fused features are fed into a cross-branch BiFPN module for multi-scale feature interaction, incorporating both Coordinate Attention (CA) and Self-Attention (SA) to further enhance the representational power of key regions. Simultaneously, a hierarchically ordered supervision mechanism is introduced to optimize the model's discriminative capability regarding continuous variations in maturity levels. The specific implementation details for each core module are outlined below.

2.4.1. LAB Dual-Branch Gated Fusion and Improved Cross-Branch BiFPN

To simultaneously encode the luminance-driven structural details and the chrominance-driven maturity variations in oil palm fruit images, this study constructs an LAB dual-branch gated fusion module combined with an improved Cross-Branch BiFPN. As shown in Figure 5, the proposed module consists of three main stages: input adaptation and dual-branch feature extraction, scale-wise gated fusion, and improved cross-branch multi-scale feature interaction.

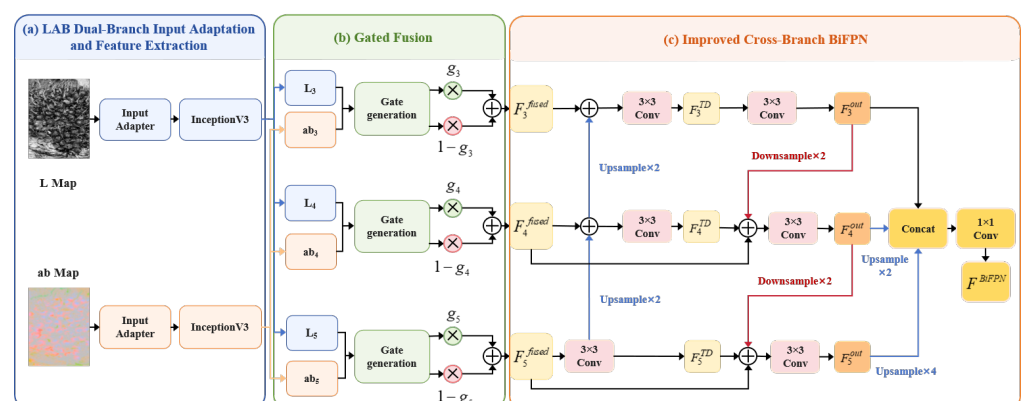


Figure 5. Architecture of the proposed LAB dual-branch gated fusion and improved Cross-Branch BiFPN module.

The input RGB image is first converted into the LAB color space and decomposed into the L map and the ab map. The L map contains one luminance channel, which mainly

describes brightness, contour, texture, and structural information. The ab map contains two chrominance channels, which mainly represent color distribution, hue variation, and maturity-sensitive chromatic information. Since the L map and ab map have different channel numbers from the three-channel input required by the InceptionV3 backbone, an input adaptation module is introduced before feature extraction.

Specifically, the input adapter consists of a 1×1 convolution, batch normalization, and ReLU activation, which performs learnable channel projection rather than simple channel replication:

$$\tilde{X}_L = \delta(\text{BN}(\text{Conv}_{1 \times 1}(X_L))) \quad (1)$$

$$\tilde{X}_{ab} = \delta(\text{BN}(\text{Conv}_{1 \times 1}(X_{ab}))) \quad (2)$$

Here, $X_L \in \mathbb{R}^{H \times W \times 1}$ and $X_{ab} \in \mathbb{R}^{H \times W \times 2}$ denote the original L and ab inputs, respectively. $\tilde{X}_L \in \mathbb{R}^{H \times W \times 3}$ and $\tilde{X}_{ab} \in \mathbb{R}^{H \times W \times 3}$ denote the adapted three-channel representations. $\text{Conv}_{1 \times 1}(\cdot)$ denotes the 1×1 convolution operation, $\text{BN}(\cdot)$ denotes batch normalization, and $\delta(\cdot)$ denotes the ReLU activation function.

Different from simply copying the L or ab channels to satisfy the input requirement of the backbone, the proposed input adapter learns modality-specific channel mappings for different LAB components. For the L branch, it enhances luminance-related structural cues, such as brightness distribution, contour, and texture. For the ab branch, it preserves and re-embeds chrominance-related cues, such as hue variation and maturity-sensitive color differences. Therefore, the input adapter not only solves the channel mismatch problem but also provides an initial feature calibration for subsequent dual-branch representation learning.

After input adaptation, $\tilde{X}_L$ and $\tilde{X}_{ab}$ are fed into two parallel InceptionV3 backbones to extract hierarchical multi-scale features. The luminance and chrominance branches generate three-scale feature sets:

$$L = \{F_3^L, F_4^L, F_5^L\}, \quad AB = \{F_3^{ab}, F_4^{ab}, F_5^{ab}\} \quad (3)$$

Here, L and AB represent the multi-scale feature sets extracted from the luminance branch and the chrominance branch, respectively. F_s^L denotes the feature map extracted from the L branch at scale s , while F_s^{ab} denotes the feature map extracted from the ab branch at scale s . The scale index $s \in \{3, 4, 5\}$ corresponds to the spatial resolutions of $1/8$, $1/16$, and $1/32$, respectively. Therefore, F_3^L and F_3^{ab} represent shallow high-resolution features, F_4^L and F_4^{ab} represent middle-level structural features, and F_5^L and F_5^{ab} represent deep semantic features.

Before gated fusion, a scale-wise feature projection is further applied to ensure that the L-branch and ab-branch features have consistent channel dimensions at each scale:

$$\hat{F}_s^L = \text{Conv}_{1 \times 1}(F_s^L), \quad \hat{F}_s^{ab} = \text{Conv}_{1 \times 1}(F_s^{ab}) \quad (4)$$

Here, $s \in \{3, 4, 5\}$ denotes the feature scale. $\hat{F}_s^L$ and $\hat{F}_s^{ab}$ denote the channel-aligned features of the L branch and ab branch at scale s , respectively. The scale-wise feature projection is used to align the channel dimensions of the multi-scale L and ab features before gated fusion.

To adaptively balance the contributions of luminance-structural features and chrominance-discriminative features, a scale-wise gated fusion mechanism is designed. For each scale, the gate weight is generated according to the concatenated L and ab features:

$$g_s = \sigma\left(\text{FC}\left(\text{GAP}\left(\text{Concat}\left(\hat{F}_s^L, \hat{F}_s^{ab}\right)\right)\right)\right) \quad (5)$$

Here, g_s denotes the learnable gate weight at scale s , $\text{Concat}(\cdot)$ denotes channel-wise concatenation, $\text{GAP}(\cdot)$ denotes global average pooling, $\text{FC}(\cdot)$ denotes the fully connected layer, and $\sigma(\cdot)$ denotes the sigmoid activation function. The value of g_s is constrained within $(0, 1)$ and represents the contribution weight of the L-branch feature. Correspondingly, $1 - g_s$ represents the contribution weight of the ab-branch feature.

The gated fusion process generates the scale-wise fused feature as follows:

$$F_s^{fused} = g_s \cdot \hat{F}_s^L + (1 - g_s) \cdot \hat{F}_s^{ab} \quad (6)$$

where F_s^{fused} denotes the fused feature at scale s . Through this gated fusion strategy, the model can dynamically adjust the relative importance of luminance and chrominance features at different scales. Compared with direct summation or simple concatenation, the gated fusion mechanism reduces redundant information interference and strengthens the complementary representation of structural and color cues.

The scale-wise gated fusion produces the following fused feature pyramid:

$$F^{fused} = \{F_3^{fused}, F_4^{fused}, F_5^{fused}\} \quad (7)$$

The fused feature pyramid is then fed into the improved Cross-Branch BiFPN for bidirectional multi-scale feature interaction. The top-down pathway transfers high-level semantic information from deep layers to shallow layers through progressive upsampling:

$$F_5^{TD} = \phi(F_5^{fused}) \quad (8)$$

$$F_4^{TD} = \phi(F_4^{fused} + \text{UP}(F_5^{TD})) \quad (9)$$

$$F_3^{TD} = \phi(F_3^{fused} + \text{UP}(F_4^{TD})) \quad (10)$$

where F_s^{TD} denotes the output feature of the top-down pathway at scale s , $\text{UP}(\cdot)$ denotes the upsampling operation, and $\phi(\cdot)$ denotes a 3×3 convolution followed by batch normalization and a nonlinear activation function. Through the top-down pathway, deep semantic information is progressively propagated to higher-resolution feature maps, thereby supplementing shallow features with stronger semantic cues.

The bottom-up pathway further propagates shallow detailed information back to deeper layers and refines the multi-scale feature representation:

$$F_3^{out} = \phi(F_3^{TD}) \quad (11)$$

$$F_4^{out} = \phi(F_4^{fused} + F_4^{TD} + \text{Down}(F_3^{out})) \quad (12)$$

$$F_5^{out} = \phi(F_5^{fused} + F_5^{TD} + \text{Down}(F_4^{out})) \quad (13)$$

where F_s^{out} denotes the output feature of the improved Cross-Branch BiFPN at scale s , and $\text{Down}(\cdot)$ denotes the downsampling operation. The bottom-up pathway enhances the interaction between shallow texture details and deep semantic representations, enabling the network to obtain more discriminative multi-scale features.

Finally, the three output features are aligned to a common spatial resolution and concatenated to generate the final fused representation:

$$F_{BIFPN} = \text{Conv}_{1 \times 1}(\text{Concat}[F_3^{out}, \text{UP}(F_4^{out}), \text{UP}(F_5^{out})]) \quad (14)$$

where F_{BiFPN} denotes the final fused feature generated by the proposed module. Since F_3^{out} has the highest spatial resolution, F_4^{out} and F_5^{out} are upsampled to the same spatial resolution as F_3^{out} before concatenation. The 1×1 convolution is then used to compress and refine the concatenated multi-scale features. The obtained F_{BiFPN} is subsequently fed into the attention enhancement module for further feature refinement.

Compared with conventional BiFPN structures that mainly aggregate multi-scale features from a single backbone, the proposed module is specifically designed for the LAB dual-branch architecture. First, the input adapter performs learnable channel mapping for the L and ab components, enabling different LAB components to better adapt to the InceptionV3 backbone. Then, the scale-wise gated fusion mechanism adaptively integrates luminance-structural and chrominance-discriminative features at each scale. Finally, the improved Cross-Branch BiFPN performs top-down and bottom-up multi-scale interaction on the fused feature pyramid. Therefore, the proposed module enhances both cross-branch complementarity and cross-scale semantic interaction, providing more discriminative representations for fine-grained oil palm fruit maturity classification.

Remark 2. Existing LAB dual-branch methods typically separate luminance and chrominance information before feature extraction, but they often lack effective channel adaptation and adaptive fusion between the two branches. It is worth noting that the proposed Input Adaptation module introduces learnable channel mapping for the L-channel and ab-channels, enabling different LAB components to better adapt to the backbone network. Furthermore, the Gated Fusion module incorporates the relative contributions of luminance-structural and chromaticity-discriminative features into the fusion process, thereby improving the complementarity of dual-branch features. This is further combined with a cross-branch BiFPN module to promote multi-scale feature interaction and enhance the discriminative representation of subtle maturity-related color and structural variations.

2.4.2. Multi-Level Attention Enhancement Mechanism

This subsection introduces the Coordinate Attention (CA) module and the Self-Attention (SA) module, which are applied sequentially for local spatial enhancement and global contextual modeling.

(1) Coordinate Attention Module In this section, the multi-level attention enhancement mechanism consists of two sequential modules: the Coordinate Attention (CA) module and the Self-Attention (SA) module.

Let the fused feature be denoted as $F^{BiFPN} \in \mathbb{R}^{H \times W \times C}$. It is then fed into the coordinate attention (CA) module to perform direction-aware weight calibration. By encoding precise spatial information, the CA module enables the network to capture both positional and channel dependencies across feature maps, thereby achieving accurate localization of key regions in oil palm fruits.

The CA module encodes global spatial information into precise coordinate dependencies along the X-axis (height) and Y-axis (width) through two unidirectional pooling operations. Global average pooling along the X- and Y-axes generates height-aggregated features z^h and width-aggregated features z^w :

$$z_c^h(h) = \frac{1}{W} \sum_{0 \leq j < W} F_c^{BiFPN}(h, j) \quad (15)$$

$$z_c^w(w) = \frac{1}{H} \sum_{0 \leq i < H} F_c^{BiFPN}(i, w) \quad (16)$$

As illustrated in Figure 6, the CA module first aggregates spatial information along the height and width directions, and then generates direction-aware attention weights through feature concatenation, transformation, and decomposition.

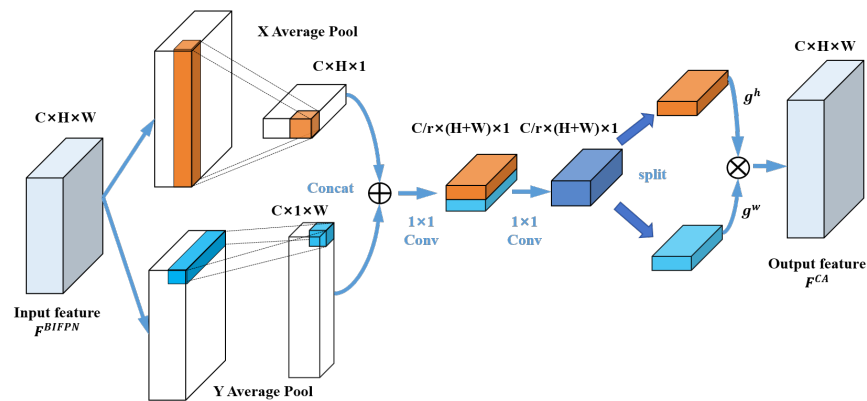


Figure 6. Structure of the Coordinate Attention (CA) module for direction-aware feature recalibration.

These aggregated features are concatenated and passed through a 1×1 convolution transformation F_1 to generate intermediate features f . Feature f is then decomposed and processed via 1×1 convolution F_h and F_w to obtain the final horizontal weights g^h and vertical weights g^w .

$$f = \delta(\text{Conv}_{1 \times 1}(\text{Concat}(z^h, z^w))) \quad (17)$$

The intermediate feature f is then split into f^h and f^w along the spatial dimension, where f^h and f^w correspond to the height and width directions, respectively.

$$g^h = \sigma(\text{Conv}_{1 \times 1}(f^h)); g^w = \sigma(\text{Conv}_{1 \times 1}(f^w)) \quad (18)$$

Here, $\delta(\cdot)$ represents the ReLU activation function. The final calibrated feature, F^{CA} , is derived by performing element-wise multiplication of F^{BIFPN} with g^h and g^w , thereby achieving feature enhancement that simultaneously accounts for channel information and precise spatial localization.

The integration of the CA module ensures that the input features for the spatial attention modeling stage have already undergone precise localization and focusing of structural information along key spatial directions.

(2) Self-Attention Module Following the completion of cross-branch feature fusion and 1D coordinate attention enhancement, the features F^{CA} obtained by the model already exhibit strong spatial saliency; however, they still lack the capacity to model global contextual dependencies. Consequently, a Self-Attention (SA) module is introduced to process the F^{CA} features, aiming to deeply mine the global, non-local dependencies within the feature maps, thereby serving as an effective complement to the local spatial focus provided by the CA mechanism.

As illustrated in Figure 7, the SA module projects the input feature into Query (Q), Key (K), and Value (V) spaces through 1×1 convolution operations, and then computes global attention responses based on the interaction between Q and K. The attention map A is computed as follows:

$$A = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \quad (19)$$

where Q, K, and V denote the query, key, and value feature matrices generated from F^{CA} , respectively; A denotes the attention weight matrix; and d_k denotes the channel dimension of the key vector, which is used as a scaling factor to stabilize the dot-product attention.

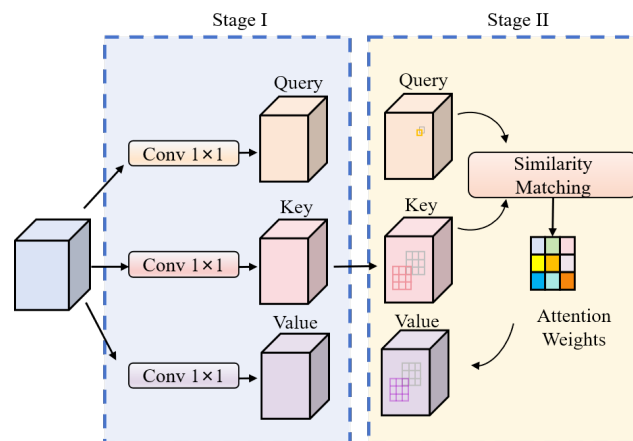


Figure 7. Structure of the Self-Attention (SA) module for global contextual dependency modeling.

The attention matrix A is then used to perform weighted summation on V to generate the attention output. The final self-attention output F^{SA} is combined with the coordinate attention feature F^{CA} through a residual connection to achieve progressive feature enhancement:

$$\begin{aligned} F^{SA} &= F^{CA} + \text{Attention}(Q, K, V) \\ &= F^{CA} + AV \end{aligned} \quad (20)$$

The integration of the SA module enables CMF-Net to effectively transcend the locality constraints of traditional convolutions, encoding long-range spatial contextual information into local features and significantly enhancing the ultimate discriminative performance of the fused features. The multi-level attention enhancement algorithm is presented in Algorithm 1.

Algorithm 1 Multi-level Attention Enhancement and Joint Training Algorithm

- 1: **Input:** BiFPN fused feature F_{BiFPN} , color distribution labels $\bar{a}, \bar{b}$
 - 2: **Output:** Enhanced feature F^{SA} , total training loss L_{total}
 - 3: Initialize regression weight λ , number of attention heads n_{head} , and training hyperparameters.
 - 4: Perform horizontal and vertical global average pooling on F_{BiFPN} according to Equations (15) and (16), to capture long-range spatial positional information.
 - 5: Apply coordinate attention weights to the original features based on Equations (17) and (18), and obtain the channel-enhanced feature F^{CA} .
 - 6: Apply linear transformation on F^{CA} to generate query Q , key K , and value V .
 - 7: Compute the correlation matrix between Q and K according to Equation (19), and apply it to V to encode long-range contextual information into the feature representation.
 - 8: Perform residual connection according to Equation (20) to complete feature enhancement, and obtain the final feature F^{SA} .
 - 9: Compute the ordinal classification loss according to Equation (22), compute the auxiliary color regression loss according to Equation (24), and update model parameters using the joint loss defined in Equation (25).
-

Remark 3. In the existing literature, the simultaneous modeling of precise spatial localization and long-range contextual dependencies has not yet gained sufficient research attention. In this paper, a multi-level attention enhancement mechanism combining CA and SA is established on the fused features, where CA strengthens direction-aware local spatial perception, while SA captures global non-local dependencies among feature regions. This design improves the model's ability to distinguish subtle maturity differences under complex background and illumination conditions.

2.5. Combined Loss of Rank-Ordered Supervision and Color-Assisted Regression

Oil palm fruit ripeness exhibits an inherent ordinal characteristic. Traditional softmax-based classification treats each category as an independent discrete label, which makes it difficult to effectively model the ordinal relationships between adjacent ripeness levels. To address this issue, an ordinal supervision mechanism based on CORAL (Consistent Rank

Logits) is introduced. In addition, an auxiliary color regression task based on color statistics is incorporated to construct a joint loss function. Specifically, a K -class ripeness classification problem is decomposed into $K - 1$ ordered binary classification tasks. As illustrated in Figure 8, the ordinal supervision branch models ripeness ranking by decomposing multi-class prediction into ordered binary decisions.

Given a ground-truth label $y \in 0, 1, \dots, K - 1$, an ordinal target vector $\mathbf{t} = [t_0, t_1, \dots, t_{K-2}]$ is constructed as:

$$t_k = \begin{cases} 1, & y > k, \\ 0, & y \leq k, \end{cases} \quad k = 0, 1, \dots, K - 2. \quad (21)$$

The network outputs the corresponding sigmoid probabilities through the ordinal prediction head, $\mathbf{p} = [p_0, p_1, \dots, p_{K-2}]$, and is optimized using the binary cross-entropy loss:

$$L_{\text{ord}} = - \sum_{k=0}^{K-2} [t_k \log p_k + (1 - t_k) \log(1 - p_k)]. \quad (22)$$

where p_k is the predicted probability of the k -th ordered binary classifier, and t_k is the corresponding ordinal target.

$$\hat{y} = \sum_{k=0}^{K-2} \mathbb{I}(p_k > 0.5), \quad (23)$$

where $\mathbb{I}(\cdot)$ is the indicator function, which equals 1 if the condition is true and 0 otherwise.

To improve the model's sensitivity to chromatic variations in oil palm fruits, an auxiliary regression branch is introduced to constrain the statistical features of the ab channels. The corresponding color regression loss is defined as:

$$L_{\text{color}} = \frac{1}{2} [(\hat{a} - \bar{a})^2 + (\hat{b} - \bar{b})^2], \quad (24)$$

where $\hat{a}$ and $\hat{b}$ denote the predicted mean values of the a and b channels, respectively. $\bar{a}$ and $\bar{b}$ denote the ground-truth mean values calculated from the ab channels of the input LAB image. This auxiliary objective provides an explicit color-sensitive supervision signal, encouraging the network to preserve small chromatic variations that are useful for distinguishing visually similar adjacent maturity stages.

The final training objective is then defined as the joint optimization of the ordinal classification loss and the auxiliary color regression loss:

$$L_{\text{total}} = L_{\text{ord}} + \lambda L_{\text{color}}, \quad (25)$$

where L_{total} denotes the total training loss, L_{ord} denotes the ordinal classification loss, L_{color} denotes the auxiliary color regression loss, and λ is the weighting coefficient used to balance the two loss terms.

Remark 4. In existing works, cross-entropy loss and its variants are commonly used for fruit maturity classification. However, they usually regard different maturity stages as independent labels and ignore the continuous ordinal relationship among adjacent categories. The joint loss mechanism proposed in this paper combines rank-ordered supervision with color-assisted regression, thus constraining the model from both maturity ranking and chromatic statistics. It can reduce misclassification between adjacent ripeness categories and improve the model's discriminative performance. In this way, CMF-Net does not rely only on categorical boundaries, but also learns the continuous color transition process underlying fruit ripening.

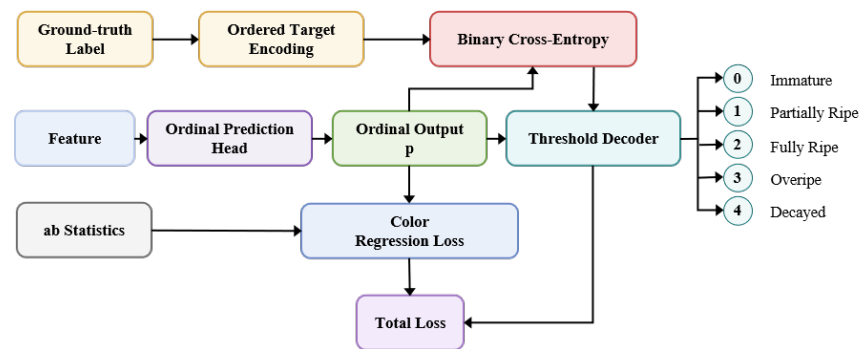


Figure 8. Structure of the joint supervision branch based on rank-ordered classification and color-assisted regression.

3. Results

3.1. Experimental Environment and Parameter Settings

All experiments were conducted under a unified deep learning environment on a GPU workstation. To ensure reproducibility, the main software environment, hardware configuration, training hyperparameters, and overfitting-control strategies are summarized in Table 2.

Table 2. Training parameter settings and overfitting control strategies.

Parameter	Setting	Parameter	Setting
Operating system	Ubuntu 20.04	Programming language	Python 3.8
Deep learning framework	PyTorch 2.0.0	GPU acceleration	CUDA 11.8
GPU	NVIDIA GeForce RTX 4090	Input image size	224 × 224
Train–validation split	8:2	Batch size	32
Training epochs	100	Optimizer	Adam
Initial learning rate	1×10^{-4}	Learning rate scheduler	ReduceLROnPlateau
Learning rate reduction factor	0.5	Scheduler patience	5 epochs
Minimum learning rate	1×10^{-6}	Model selection criterion	Best validation accuracy
Dropout rates	0.5, 0.4, 0.3	Label smoothing factor	0.02
Auxiliary color loss weight	0.1	Validation augmentation	Not used

For each dataset, the same train–validation split and preprocessing strategy were used for CMF-Net and all baseline models to ensure a fair comparison. The training and validation sets were divided at a ratio of 8:2, and all input images were resized to 224 × 224 pixels. The model checkpoint with the highest validation accuracy was retained for final evaluation. To reduce overfitting, data augmentation was applied only to the training set, while the validation set was kept unchanged. Dropout regularization, label smoothing, auxiliary color supervision, and dynamic learning-rate adjustment were also adopted. The main training hyperparameters and overfitting-control settings are summarized in Table 2.

3.2. Experimental Evaluation Metrics

To accurately and comprehensively characterize the results of data processing, this study employs Accuracy, Precision, Recall, Specificity, and the F1-score as comprehensive evaluation metrics. All of these metrics are constructed upon the foundation of a confusion matrix; by performing calculations based on the classification outcomes of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), they reflect, from various perspectives, the model's classification accuracy and robustness within complex contexts.

3.3. Analysis of Comparison Results

To comprehensively evaluate the effectiveness of the proposed CMF-Net for oil palm fruit maturity classification, comparative experiments were conducted with both classical convolutional neural networks and recent advanced visual recognition models. The selected comparison models include VGG16 [32], MobileNetV2 [33], DenseNet121 [34], Improved CNN [35], ResNet152V2 [36], MobileViT-CBAM [37], Swin-T [38], ConvNeXt-Tiny [39], and EfficientNetV2-S [40]. These models cover traditional CNN architectures, lightweight networks, attention-enhanced models, and recent Transformer- or ConvNeXt-style architectures, thereby providing a more comprehensive benchmark for evaluating the proposed method.

All comparison models were trained and evaluated under the same dataset partition and experimental environment. For fairness, the input image size was fixed at 224×224 , and the same training and validation sets were used for all models. In addition to classification metrics, model complexity was also reported, including the number of parameters, FLOPs, and single-image inference time. The validation loss and validation accuracy curves are shown in Figure 9.

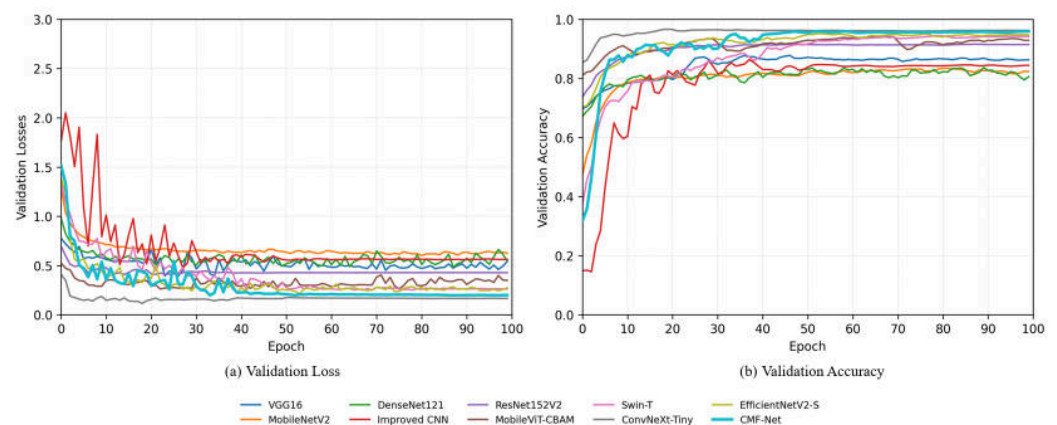


Figure 9. (a) Change in Model Validation Loss; (b) Change in Model Validation Accuracy.

As shown in Figure 9, the traditional CNN-based models, such as VGG16, MobileNetV2, DenseNet121, Improved CNN, and ResNet152V2, show relatively limited performance in this fine-grained maturity classification task. This indicates that directly learning from RGB images may be insufficient for capturing subtle color transitions between adjacent oil palm fruit maturity stages. Recent models such as Swin-T, ConvNeXt-Tiny, and EfficientNetV2-S achieve stronger recognition performance, demonstrating the benefit of advanced feature modeling strategies. However, CMF-Net maintains highly competitive and stable validation performance, suggesting that the task-specific LAB dual-branch design is effective for maturity-related color and structural feature representation.

The quantitative comparison results are presented in Table 3. CMF-Net achieves the best Accuracy, Recall, and F1-score among all compared models, reaching 0.9588, 0.9666, and 0.9620, respectively. Compared with the classical CNN baselines, CMF-Net shows a clear improvement across the main classification metrics. Compared with recent strong models, including Swin-T, ConvNeXt-Tiny, and EfficientNetV2-S, CMF-Net still obtains the highest overall classification accuracy and F1-score. This result indicates that the proposed model is particularly effective in reducing confusion between adjacent maturity categories, where subtle chromatic differences are often difficult to distinguish.

Table 3. Performance and complexity comparison of different classification models on the main dataset.

Model	Params (M)	FLOPs (G)	Infer. (ms)	Acc. (%)	Prec. (%)	Recall (%)	Spec. (%)	F1 (%)	Train Acc.	Val. Acc.
VGG16 [32]	14.85	30.71	3.13	87.54	89.53	89.32	96.57	89.36	93.97	85.85
MobileNetV2 [33]	2.42	0.60	2.78	81.78	81.78	82.78	93.81	81.72	91.59	81.84
DenseNet121 [34]	7.05	5.67	3.11	83.32	83.45	83.32	93.72	83.32	93.44	83.74
Improved CNN [35]	0.49	1.48	2.76	87.96	88.72	87.96	95.61	87.97	93.58	84.47
ResNet152V2 [36]	58.59	22.81	4.20	89.55	89.57	89.55	96.06	89.55	99.55	91.43
MobileViT-CBAM [37]	3.00	0.60	2.99	90.71	91.01	90.71	97.39	90.71	99.60	93.35
Swin-T [38]	27.52	4.37	0.60	95.14	95.66	96.40	98.68	96.16	99.97	94.15
ConvNeXt-Tiny [39]	27.82	4.45	0.49	95.42	97.11	96.35	99.00	96.11	99.98	96.02
EfficientNet V2-S [40]	20.18	2.71	0.32	94.65	95.70	95.40	98.62	95.52	99.71	94.36
CMF-Net	26.74	5.37	0.84	95.88	96.02	96.66	98.91	96.20	100.00	95.98

In terms of computational complexity, CMF-Net still has certain limitations. Owing to the use of dual-branch feature extraction and multi-level feature fusion, its parameter size and FLOPs are higher than those of lightweight models such as MobileNetV2, MobileViT-CBAM, and EfficientNetV2-S. This indicates that further optimization is still required before deployment on highly resource-constrained devices. Nevertheless, compared with heavier CNN architectures such as VGG16 and ResNet152V2, CMF-Net requires fewer FLOPs and achieves a shorter inference time, while obtaining higher accuracy, recall, and F1-score. More importantly, CMF-Net is designed for fine-grained oil palm fruit maturity recognition, where subtle color transitions and adjacent-stage confusion are the main challenges. Therefore, the increased computational cost is accompanied by improved maturity-discriminative capability. In future work, lightweight optimization strategies, such as model pruning, knowledge distillation, quantization, and lightweight backbone replacement, will be explored to further reduce the computational burden and improve the suitability of CMF-Net for mobile terminals, embedded devices, and field-side agricultural inspection systems. The performance gain of CMF-Net can be attributed to its collaborative design. First, the LAB color space separates luminance and chrominance information, allowing the model to learn structural and color-related cues through different branches. Second, the gated fusion and BiFPN modules promote adaptive cross-branch and multi-scale feature interaction. Third, the coordinate attention and self-attention modules enhance local key-region perception and global contextual modeling. Finally, the auxiliary color regression constraint and ordinal supervision further improve the model's sensitivity to continuous maturity changes. These components jointly enable CMF-Net to achieve stronger discriminative ability for oil palm fruit maturity classification.

3.4. Ablation Study

Based on the ablation study results presented in Table 4, it is evident that each improved module contributes positively to enhancing the performance of CMF-Net. Compared to the baseline model, which did not incorporate any enhancement strategies, the introduction of the Input Adapter resulted in significant improvements in both model accuracy and F1 score, demonstrating that learnable input mapping effectively enhances dual-branch feature representation. Specifically, the accuracy increased from 87.95% to 91.54%, and the F1 score increased from 87.01% to 91.84%.

Building upon this foundation, the Gated Fusion and BiFPN structures further bolstered the capability for cross-modal, multi-scale feature fusion, leading to sustained improvements in model performance. After adding the Gated Fusion module, the accuracy increased to 92.28%, and the precision increased from 92.29% to 93.64%. With the further integration of BiFPN, the accuracy increased to 93.66%, the recall increased from 91.48% to 93.84%, and the F1 score increased from 92.04% to 94.20%.

With the integration of the CA and SA attention mechanisms, the model's capacity to capture information regarding key regions and global contextual relationships was enhanced, resulting in a steady increase in the F1 score. After introducing CA, the accuracy and F1 score increased to 94.19% and 94.42%, respectively. After further adding SA, the accuracy, recall, and F1 score reached 94.58%, 94.86%, and 94.69%, respectively.

Finally, the incorporation of rank-ordered supervision into the complete model yielded optimal results across all evaluation metrics; this indicates that ordered supervision effectively captures the continuous, ordinal nature of oil palm fruit maturity levels and mitigates confusion between adjacent categories, thereby achieving superior classification performance. The complete model achieved an accuracy of 95.88%, precision of 96.02%, recall of 96.66%, specificity of 98.91%, and F1 score of 96.20%. Compared with the baseline model, the accuracy, precision, recall, specificity, and F1 score increased by 7.93, 9.35, 8.32, 2.04, and 9.19 percentage points, respectively.

Table 4. Ablation study.

Model	Input Adapter	Gated Fusion	BiFPN	CA	SA	Ordinal Loss	Acc. (%)	Prec. (%)	Recall (%)	Spec. (%)	F1 (%)
Baseline	×	×	×	×	×	×	87.95	86.67	88.34	96.87	87.01
Model1	✓	×	×	×	×	×	91.54	92.29	90.90	97.69	91.84
Model2	✓	✓	×	×	×	×	92.28	93.64	91.48	97.87	92.04
Model3	✓	✓	✓	×	×	×	93.66	94.65	93.84	98.26	94.20
Model4	✓	✓	✓	✓	×	×	94.19	95.18	94.60	96.60	94.42
Model5	✓	✓	✓	✓	✓	×	94.58	94.54	94.86	98.43	94.69
Ours	✓	✓	✓	✓	✓	✓	95.88	96.02	96.66	98.91	96.20

3.5. Ablation Study on the Weight Coefficient λ

To determine an appropriate value for the auxiliary loss weight coefficient λ , a sensitivity analysis is conducted under fixed network architecture and training settings. Different values of λ are evaluated, and the results are summarized in Table 5. As λ increases, the model performance exhibits a trend of first improving and then declining. When $\lambda = 0$, no auxiliary supervision is introduced, and the model achieves relatively lower performance. As λ increases to a moderate value, the auxiliary loss effectively enhances the feature learning capability of the model. In particular, when $\lambda = 0.1$, both Accuracy and F1-score reach their optimal values. However, further increasing λ introduces interference to the learning of the main task, leading to performance degradation. Therefore, in the subsequent experiments, λ is set to 0.1.

Table 5. Model performance under different values of λ .

λ	Accuracy (%)	F1-Score (%)
0.00	93.23	94.07
0.05	94.55	94.54
0.10	95.35	95.57
0.20	93.97	94.72
0.50	93.66	94.51

3.6. Cross-Dataset Generalization Experiment

To further evaluate the generalization ability of the proposed CMF-Net, additional experiments were conducted on three external datasets, including the Ripeness of Oil Palm Fruit Dataset, the Palm Ripeness Dataset, and the Date Fruit Dataset. The first two datasets are directly related to oil palm fruit maturity recognition, while the Date Fruit Dataset was used as a related palm-fruit maturity classification dataset. This design was adopted because publicly available oil palm fruit maturity datasets are relatively scarce, making it difficult to perform extensive cross-region and cross-source validation using oil palm data alone. Therefore, the external datasets were used to assess the robustness of CMF-Net from two complementary perspectives: direct oil palm fruit generalization and related palm-family fruit maturity recognition.

For each external dataset, the same training and validation protocol was adopted for all comparison models. The compared models include ResNet152V2, MobileViT-CBAM, ConvNeXt-Tiny, EfficientNetV2-S, Swin-T, and the proposed CMF-Net. The evaluation metrics include Accuracy, Precision, Recall, Specificity, and F1-score. The experimental results are shown in Table 6.

As shown in Table 6, CMF-Net demonstrates strong generalization performance across different datasets. On the Ripeness of Oil Palm Fruit Dataset, CMF-Net achieves the best results across all five evaluation metrics, with an accuracy of 98.96%, precision of 98.95%, recall of 98.94%, specificity of 99.46%, and F1-score of 98.93%. This indicates that the LAB dual-branch feature representation remains effective when the model is evaluated on an external oil palm fruit dataset.

Table 6. Cross-dataset generalization performance of different classification models.

Dataset	Model	Acc. (%)	Prec. (%)	Recall (%)	Spec. (%)	F1 (%)
Ripeness of Oil Palm Fruit	ResNet152V2	94.93	94.97	94.93	97.46	94.94
	MobileViT-CBAM	98.81	98.82	98.81	99.31	98.39
	ConvNeXt-Tiny	98.75	98.75	98.75	99.39	98.91
	EfficientNetV2-S	97.46	97.46	97.46	98.73	97.46
	Swin-T	98.51	98.54	98.51	99.24	98.50
	CMF-Net	98.96	98.95	98.94	99.46	98.93
Palm Ripeness Dataset	ResNet152V2	86.27	88.94	84.81	92.93	84.82
	MobileViT-CBAM	96.03	96.49	95.52	97.94	95.78
	ConvNeXt-Tiny	96.03	96.67	95.52	97.94	95.82
	EfficientNetV2-S	94.12	93.98	94.07	97.14	93.97
	Swin-T	96.00	94.92	96.11	97.03	95.93
	CMF-Net	96.16	94.25	96.15	97.04	95.63
Date Fruit Dataset	ResNet152V2	98.52	98.45	98.28	99.49	98.36
	MobileViT-CBAM	99.61	99.53	99.47	99.87	99.50
	ConvNeXt-Tiny	99.57	99.54	99.58	99.88	99.56
	EfficientNetV2-S	99.61	99.53	99.54	99.87	99.53
	Swin-T	99.58	99.55	99.66	99.87	99.64
	CMF-Net	99.63	99.58	99.83	99.89	99.79

On the Palm Ripeness Dataset, which contains a relatively small number of samples, CMF-Net achieves the highest accuracy and recall, reaching 96.16% and 96.15%, respectively. Although ConvNeXt-Tiny and MobileViT-CBAM obtain slightly higher precision or specificity, CMF-Net maintains the strongest overall recognition accuracy and shows stable sensitivity to maturity categories. This result suggests that the proposed model can still maintain competitive performance under limited-sample external data conditions.

For the Date Fruit Dataset, CMF-Net achieves the best performance across all evaluation metrics, with an accuracy of 99.63%, precision of 99.58%, recall of 99.83%, specificity

of 99.89%, and F1-score of 99.79%. This result indicates that the proposed LAB-based dual-branch representation is not limited to the main oil palm fruit dataset, but can also maintain strong discriminative ability when applied to a related palm-family fruit dataset with different fruit appearance and imaging conditions. The superior recall and F1-score further demonstrate that CMF-Net can effectively capture maturity-related color transitions and structural cues across different fruit varieties, suggesting its potential applicability to broader fruit maturity classification tasks.

Overall, the cross-dataset experiments demonstrate that CMF-Net not only achieves strong performance on the main Oil Palm Fruit Ripeness Dataset but also shows promising recognition performance on external datasets with different sample sizes, fruit appearances, and imaging conditions. These results indicate that the proposed LAB-based dual-branch feature representation and fusion strategy have good generalization potential for related fruit maturity classification tasks.

3.7. 5-Fold Cross-Validation

To assess the stability and reliability of the proposed CMF-Net model, a 5-fold cross-validation experiment was conducted. The results are shown in Figure 10.

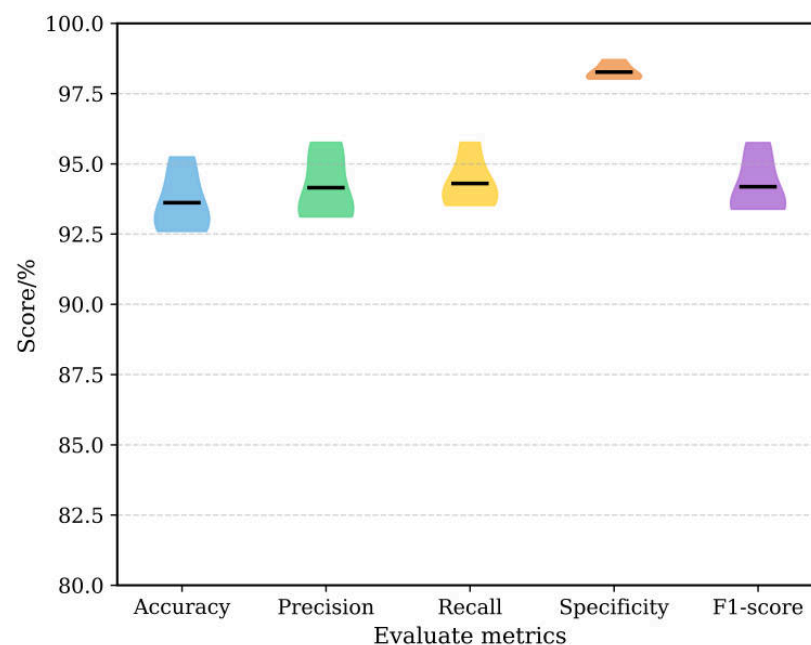


Figure 10. 5-fold cross-validation results of CMF-Net.

Overall, CMF-Net demonstrates stable performance across five evaluation metrics, including accuracy, precision, recall, specificity, and F1-score. Among them, specificity achieves the highest score, remaining around 98% with a very narrow distribution, indicating that the model has a strong ability to correctly identify negative samples and reduce false positive predictions. Accuracy, precision, recall, and F1-score are mainly distributed within the range of approximately 93% to 96%, showing relatively concentrated distributions across different folds. This suggests that CMF-Net maintains consistent classification performance under different data subset partitions. In particular, precision and recall exhibit stable distributions, indicating that the model achieves a good balance between correctly identifying mature classes and reducing misclassification. These results further validate the robustness and generalization capability of CMF-Net for oil palm fruit maturity classification.

3.8. Statistical Robustness and Significance Analysis

To further evaluate the reliability of the experimental results, a statistical robustness analysis was conducted based on five-fold cross-validation. Five representative models were selected: ResNet152V2 as a deep CNN baseline, MobileViT-CBAM as a lightweight attention-enhanced model, Swin-T as a Transformer-based model, EfficientNetV2-S as an efficient convolutional network, and ConvNeXt-Tiny as a recent high-performing ConvNet architecture. These models cover different mainstream visual recognition paradigms and provide a balanced comparison with CMF-Net.

For each model, five-fold cross-validation was performed under the same preprocessing strategy, input size, training settings, and evaluation metrics. The results are reported as mean $\pm$ standard deviation, and 95% confidence intervals were calculated to describe the stability of model performance. The p -values indicate paired comparisons with CMF-Net based on accuracy. Paired statistical tests were further conducted between CMF-Net and each representative baseline using fold-wise results.

As shown in Table 7, CMF-Net achieves the highest mean accuracy and F1-score, reaching $95.88\% \pm 0.32\%$ and $96.20\% \pm 0.35\%$, respectively. The corresponding 95% confidence intervals are [95.48, 96.28] for accuracy and [95.77, 96.63] for F1-score, indicating stable performance across different folds. Compared with ResNet152V2, MobileViT-CBAM, Swin-T, and EfficientNetV2-S, CMF-Net shows statistically significant improvements. For ConvNeXt-Tiny, the p -value is 0.087, indicating that the difference is not significant at the 0.05 level, although CMF-Net still obtains slightly higher mean accuracy and F1-score. Overall, these results indicate that CMF-Net performs consistently across different data partitions and remains competitive against representative CNN-based, Transformer-based, and modern ConvNet models.

Table 7. Statistical analysis of representative models based on five-fold cross-validation.

Model	Accuracy (%)	Acc. 95% CI	F1-Score (%)	F1 95% CI	p -Value (vs. CMF-Net)
ResNet152V2	91.26 ± 0.84	[90.22, 92.30]	91.08 ± 0.91	[89.95, 92.21]	<0.01
MobileViT-CBAM	93.42 ± 0.63	[92.64, 94.20]	93.36 ± 0.68	[92.52, 94.20]	<0.05
Swin-T	94.25 ± 0.51	[93.62, 94.88]	95.74 ± 0.46	[95.17, 96.31]	<0.05
EfficientNetV2-S	94.63 ± 0.48	[94.03, 95.23]	95.31 ± 0.52	[94.67, 95.95]	<0.05
ConvNeXt-Tiny	95.41 ± 0.39	[94.93, 95.89]	96.03 ± 0.41	[95.52, 96.54]	0.087
CMF-Net	95.88 ± 0.32	[95.48, 96.28]	96.20 ± 0.35	[95.77, 96.63]	–

3.9. Confusion Matrices and Visual Classification Analysis

3.9.1. Confusion Matrix Analysis

To meticulously evaluate the model's specific performance across various maturity categories, we computed the confusion matrix for CMF-Net on the validation set, the results are presented in Figure 11. The results show that CMF-Net achieves high recognition accuracy for most maturity categories, with most samples concentrated along the main diagonal. This is particularly evident in the two extreme categories, Immature and Decayed, where the classification accuracies reach 98.68% and 98.78%, respectively. Most misclassifications occur between adjacent categories, such as Partially Ripe and Fully Ripe, which is consistent with the continuous color transition during oil palm fruit ripening. These results indicate that subtle color differences between adjacent stages remain the main source of classification difficulty. Nevertheless, the concentration of most samples along the diagonal suggests that the proposed LAB dual-branch representation and auxiliary color supervision help alleviate adjacent-stage confusion.

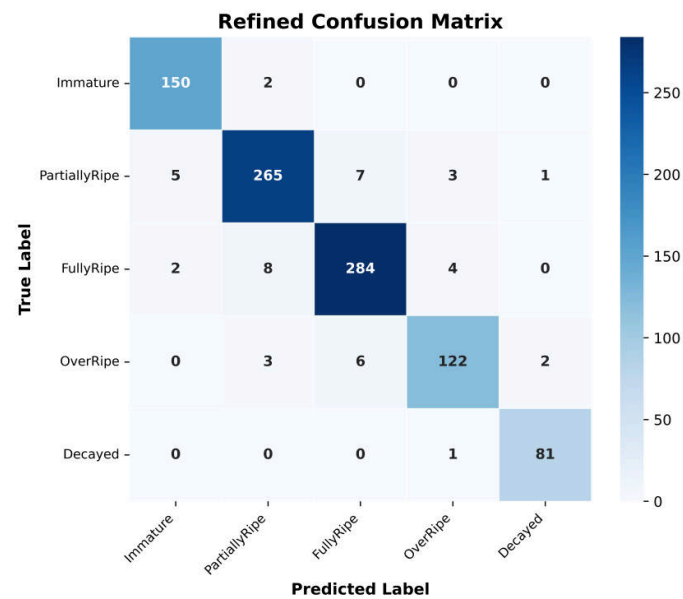


Figure 11. CMF-Net Confusion Matrix.

3.9.2. t-SNE Feature Visualization Analysis

To further analyze the discriminative ability of the learned feature representations, this study employed t-distributed stochastic neighbor embedding (t-SNE) to visualize the high-dimensional features extracted from the penultimate layer of the validation set. t-SNE is a nonlinear dimensionality reduction method that maps high-dimensional data into a low-dimensional space while preserving local neighborhood relationships as much as possible [41]. Therefore, it is widely used for visualizing feature separability in classification tasks. In this study, the extracted deep features were projected into a two-dimensional space using t-SNE, allowing the distribution and separation of different maturity categories to be intuitively observed.

The results of the t-SNE feature visualization are presented in Figure 12. Figure 12a shows that the baseline model produces noticeable overlap among several maturity categories, indicating that its feature representation is less discriminative. In contrast, CMF-Net, as shown in Figure 12b, presents more compact intra-class distributions and clearer inter-class separation. This indicates that the proposed feature fusion and attention enhancement strategy improves the representation of maturity-related visual information. The clearer separation between neighboring maturity categories further supports the effectiveness of CMF-Net in modeling subtle color and texture transitions.

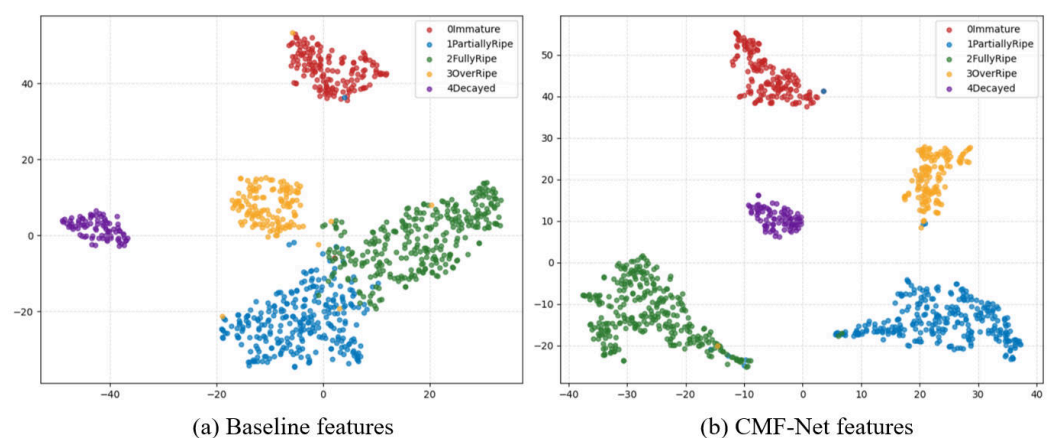


Figure 12. t-SNE Feature Visualization Analysis Results.

3.9.3. Grad-CAM Visualization Analysis

Grad-CAM technology was utilized to further analyze the model's decision-making regions. The Grad-CAM visualization results are presented in Figure 13, demonstrating that CMF-Net's high-response heatmaps precisely focus on the main body of the fruit clusters and their color-rich areas, thereby effectively suppressing interference from background soil and foliage. This result corroborates the t-SNE analysis, indicating that the model not only achieves effective separation within the feature space but also accurately localizes key color features within the physical space, thereby exhibiting exceptional interference resistance and robustness.

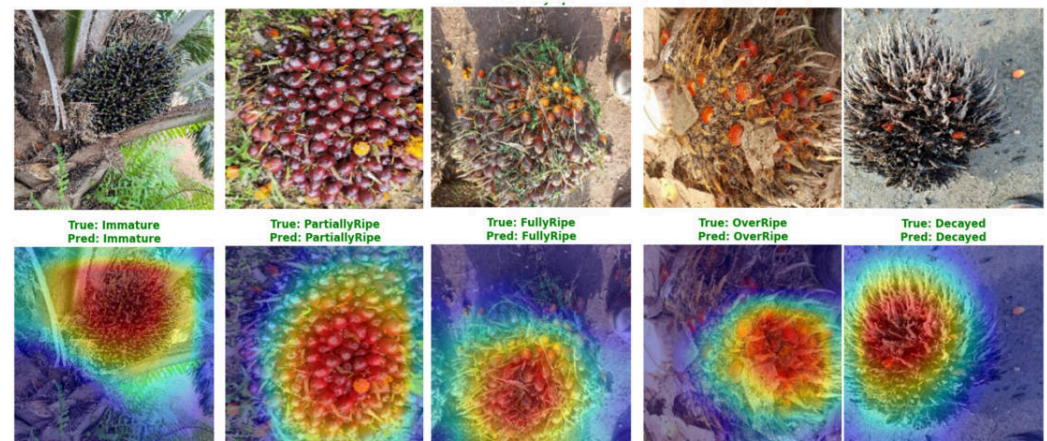


Figure 13. Grad-CAM Attention Region Visualization Results.

4. Discussion

The experimental results show that CMF-Net achieves competitive performance in oil palm fruit maturity classification compared with both conventional CNN-based models and recent advanced visual recognition models. Traditional CNN architectures, such as VGG16, DenseNet121, MobileNetV2, and ResNet152V2, mainly learn features from RGB images in a single-stream manner, which may limit their ability to distinguish adjacent maturity stages with gradual color variations. In contrast, CMF-Net introduces LAB-based luminance–chrominance decomposition, allowing structural information and color-related maturity cues to be learned through different branches.

Compared with recent models such as MobileViT-CBAM, Swin-T, ConvNeXt-Tiny, and EfficientNetV2-S, CMF-Net also shows strong overall performance. Although these models benefit from attention mechanisms, Transformer-style representation, or modernized convolutional structures, CMF-Net obtains the best accuracy, recall, and F1-score, reaching 95.88%, 96.66%, and 96.20%, respectively. This indicates that a task-specific design based on color-space decomposition and feature fusion is effective for fruit maturity classification.

The advantage of CMF-Net mainly comes from the coordinated modeling of luminance, chrominance, multi-scale features, and ordinal maturity relationships. The LAB dual-branch structure separates brightness-related structural information and color-related maturity cues, while the gated fusion and BiFPN modules enhance cross-branch and multi-scale feature interaction. In addition, the CA and SA modules improve regional feature perception and global contextual modeling. The auxiliary color regression and ordinal supervision further help the model preserve subtle chromatic variations and reduce confusion between neighboring maturity categories.

The cross-dataset experiments further demonstrate the generalization potential of the proposed method. CMF-Net achieves strong performance on external oil palm fruit datasets and also shows promising results on the Date Fruit Dataset. These results suggest that the proposed LAB-based feature representation has potential transferability to related fruit maturity classification tasks with different sample sizes, fruit appearances, and imaging conditions.

Despite these advantages, several limitations should be acknowledged. First, publicly available oil palm fruit maturity datasets are still limited, and the current experiments cannot fully cover variations caused by different regions, cultivars, seasons, imaging devices, and field environments. Second, the dual-branch structure of CMF-Net increases computational cost compared with lightweight models. Although CMF-Net achieves a favorable accuracy–complexity trade-off, further optimization is still needed for mobile or low-power embedded deployment.

In practical applications, CMF-Net is more suitable for scenarios where recognition accuracy is prioritized, such as automated grading systems, quality inspection terminals, and harvesting decision-support platforms. Future work will focus on collecting larger-scale multi-region and multi-device datasets, improving model efficiency through pruning, knowledge distillation, backbone replacement, and quantization, and extending the framework from image-level classification to real-time fruit detection and maturity assessment in field scenes.

5. Conclusions

This study proposed CMF-Net, an LAB dual-branch feature fusion network for five-class oil palm fruit maturity classification. By decomposing input images into luminance and chrominance components, the model separately extracts structural information and color-related maturity cues. The input adaptation module, gated feature fusion, Cross-Branch BiFPN, and multi-level attention mechanism enhance cross-branch complementarity and multi-scale feature interaction. In addition, ordinal supervision and color-assisted regression are introduced to model the progressive relationship among maturity stages. From a methodological perspective, CMF-Net provides a task-oriented framework that integrates color-space decomposition, adaptive feature fusion, and maturity-order modeling for image-based fruit maturity recognition.

Experimental results on the main Oil Palm Fruit Ripeness Dataset show that CMF-Net achieves an accuracy of 95.88%, a precision of 96.02%, a recall of 96.66%, a specificity of 98.91%, and an F1-score of 96.20%. Compared with classical CNN models and recent advanced visual recognition models, including Swin-T, ConvNeXt-Tiny, and EfficientNetV2-S, CMF-Net achieves the best accuracy, recall, and F1-score. The ablation experiments further verify the contribution of the main components, including LAB decomposition, input adaptation, gated fusion, multi-scale interaction, attention enhancement, and ordinal supervision. Cross-dataset experiments and five-fold cross-validation further demonstrate the generalization potential and stability of CMF-Net. These results indicate that the proposed method has practical potential for non-destructive maturity assessment, automated grading, harvesting decision support, and quality control in oil palm production.

At its current computational scale, CMF-Net is more suitable for GPU-equipped workstations, edge servers, and industrial inspection terminals where recognition accuracy is prioritized. For deployment on embedded platforms, mobile inspection devices, and field-side agricultural equipment, further lightweight optimization is still needed. In the future, we will establish independent datasets or evaluate the proposed model under real-world production conditions. We will also focus on collecting larger-scale multi-region and multi-device oil palm fruit datasets, improving model efficiency through backbone replacement,

pruning, distillation, and quantization, and extending the framework from image-level classification to real-time fruit detection and maturity assessment in field scenes.

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